Northeastern University
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NVIDIA Comprehensive Risk Management Report

ALY6130: Risk Management for Analytics

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Table of Contents

1. Introduction **3**
2. Risk Treatment and Response Plan **3**
2.1 Company Background 3
2.2 How I Identified These Risks 4
3. Qualitative Risk Analysis & Prioritization **5**
3.1 Risk Register and Scoring 5
3.2 Why Supply Chain & Geopolitical Are "High Priority" 7
3.3 Heat Maps 7
Figure 1. NVIDIA Risks Heat Map (1–9 Scale) 8
4. Risk Scenario Modeling & Treatment Plans **8**
4.1 Influence Diagram (Focus on Geopolitical Tensions) 8
Figure 2. NVIDIA Influence Diagram 9
4.2 Scenario Tree 9
Figure 3. NVIDIA Scenario Tree Diagram 12
4.3 Detailed Treatment Plans 13
5. Quantitative Risk Analysis **14**
5.1 Data Sources 14
5.2 Monte Carlo Simulation (Summary) 14
5.3 Cumulative Distribution (CDF) 14
Figure 4. NVIDIA Cumulative Distribution Function (CDF) 15
6. Risk Management Decision Making & Key Risk Indicators **15**
6.1 Decision Making 15
6.2 Key Risk Indicators (KRIs) 15
7. Continuous Risk Management & Communication **16**
7.1 Ongoing Updates 16
7.2 Organizational Communication 16
7.3 Management's Role 16
8. Conclusion **17**
9. Reference **17**

1. Introduction

This report shows how we can manage NVIDIA's biggest risks using the full risk management process. Over the course of this class, I looked at NVIDIA's annual report, news articles, and other sources to learn about the things that can hurt NVIDIA's operations and growth. Here, I combine all the major steps (identification, assessment, mitigation, and monitoring) into one comprehensive plan.

You will see that the risks revolve around over-reliance on TSMC, geopolitical issues in Asia, strong competition from AMD and Intel, and a few other concerns. The main goal is to protect NVIDIA's revenue and reputation by planning realistic steps to reduce these dangers, track warning signs, and respond if they grow bigger. This plan also explains how we communicate these risks to different groups at NVIDIA like the supply chain team, finance department, and top executives.

2. Risk Treatment and Response Plan

2.1 Company Background

NVIDIA is known for graphics processing units (GPU) that power games, data centers, artificial intelligence, autonomous cars, healthcare innovations, and more. Its technology helps many industries grow faster. However, being a market leader means facing new threats, including:

- **Global supply chain vulnerabilities** (especially in semiconductors)
- **Government regulations** or trade restrictions (like U.S.-China chip policies)

- **Competitive pressure** from rival GPU makers

2.2 How I Identified These Risks

To find NVIDIA's key risks, I followed NASA's Continuous Risk Management (CRM) approach. According to CRM, we first define and maintain a risk management plan, then identify all possible conditions that could cause trouble for our organization. Below is a simple explanation for how to use the CRM step:

1. **Inputs From Different Sources**
    - **Form 10-K & Investor Reports:** I reviewed NVIDIA's latest 2024 annual report to see official statements on supply chain, competition, and market risks.
    - **External Press & Tech Analysis:** Articles and analyses (from sources like CNBC, Nasdaq, and Reuters) flagged potential export restrictions, new AI competitors, and geopolitical tensions.
2. **Identify Conditions That Lead to Risks**
    - Using NASA's guidelines, I looked for anything that might disrupt NVIDIA's performance requirements (like GPU production or AI growth). Over reliance on TSMC or a new U.S.-China trade policy.
3. **Draft Initial Risk Statements**
    - For each possible threat, I wrote a short statement describing the cause, result, and potential impact. For example, the risk of supply chain disruptions caused by overreliance on TSMC, resulting in production delays and lost sales.
4. **Add Risks to the Register**

- As NASA's CRM suggests, once I found a risk, I put it into the Risk Register. Each risk also has a category: operational, financial, competitive, or geopolitical to help with future analysis.

By following NASA's CRM steps, I ended up with the **six main risks** for NVIDIA's business:

1. **Supply Chain Disruptions**
2. **Export Restrictions**
3. **Strong Competition**
4. **Geopolitical Tensions**
5. **Emergence of Cost-Effective AI Models**
6. **Market Valuation Volatility**

These became the foundation for all further analysis in the next risk register.

3. Qualitative Risk Analysis & Prioritization

3.1 Risk Register and Scoring

Risk #	The Risk of/That	Caused By	Resulting In	Consequence/ Impact
1	Supply chain disruptions	Over-reliance on TSMC	Production delays and lost sales	Financial and operational harm
2	Export restrictions	U.S. government policies	Reduced sales in key markets	Revenue loss
3	Strong competition	AMD and Intel improving their products	Loss of customers and price reductions	Reduced profits and weaker leadership
4	**Geopolitical tensions**	**U.S.-China conflict and Taiwan's status**	**Disruptions in supply chain and production**	**Severe global supply chain interruptions**

5	Emergence of cost-effective AI models	Development of affordable AI by competitors	Reduced demand for NVIDIA GPUs	Loss of market share and decreased revenue
6	Market valuation volatility	Investor concerns over competitive threats	Rapid decline in stock value	Significant financial instability

Risk #	Chance Score	Impact Score	Risk Score
1 - Supply Chain Disruptions	7	8	56
2 - Export Restrictions	5	8	40
3 - Strong Competition	5	6	30
4 - Geopolitical Tensions	**7**	**9**	**63**
5 - Emerging Low-Cost AI Models	5	6	30
6 - Market Valuation Volatility	5	6	30

Risk #	Priority (H, M, L)	Mitigation	Risk Owner	Open/ Closed
1 - Supply Chain Disruptions	H	Partner with alternative suppliers like Samsung, Intel. Build local facilities	Supply Chain Manager	Open
2 - Export Restrictions	M	Build partnerships in safe markets and follow trade rules. Develop localized product versions	Global Trade Manager	Open
3 - Strong Competition	M	Increase research and innovation to stay ahead of competitors. Accelerate GPU release cycles	R&D Manager	Open
4 - Geopolitical Tensions	**H**	**Keep an eye on political tensions and diversify chip production to non-volatile regions**	**Risk Management Team**	**Open**
5 - Emerging	M	Keep creating better	Strategy Manager	Open

Low-Cost AI Models		and unique products and develop affordable AI solutions for mid-tier enterprises.		
6 - Market Valuation Volatility	M	Plan finances carefully and enhance investor transparency	Finance Manager	Open

High Priority: #1 (Supply Chain), #4 (Geopolitical)

Medium Priority: #2 (Export), #3 (Competition), #5 (Low-Cost AI), #6 (Valuation)

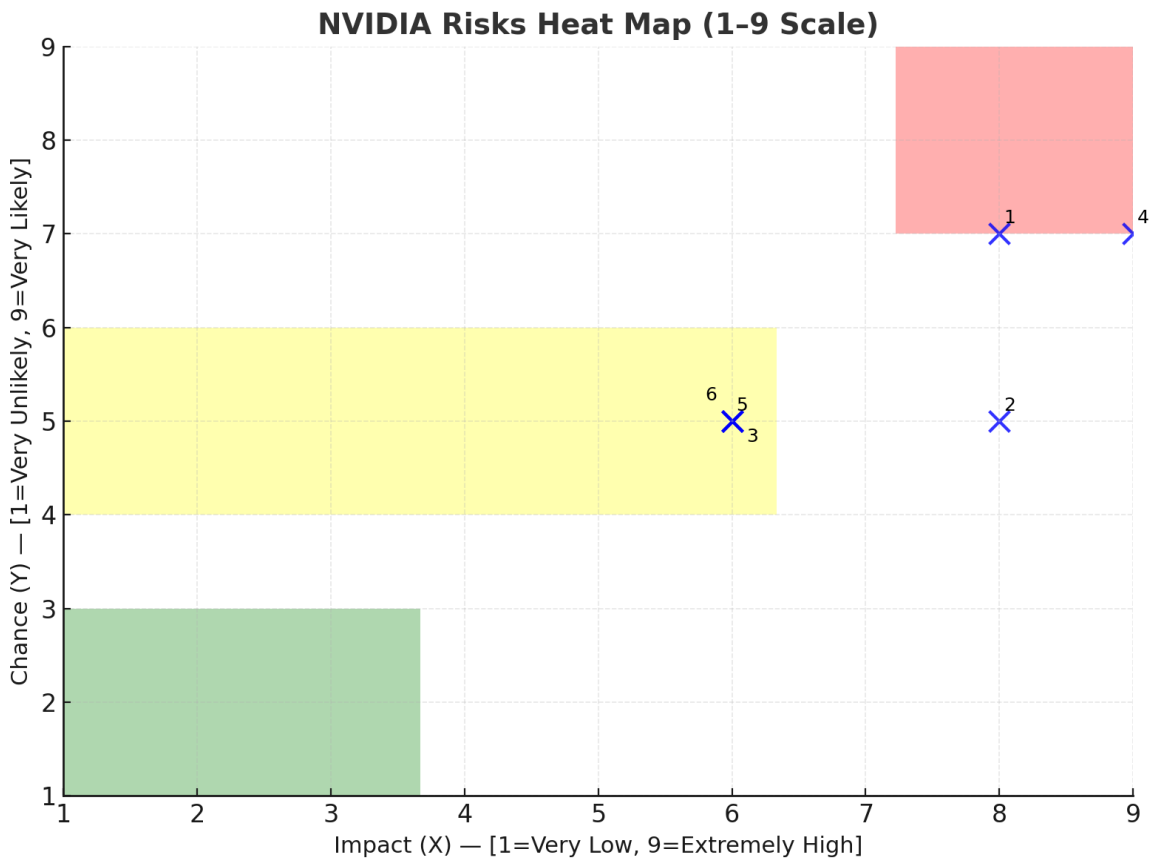
3.2 Why Supply Chain & Geopolitical Are "High Priority"

- **Supply Chain (#1):** So reliant on TSMC that if TSMC has a crisis or backlog, NVIDIA's sales could drop sharply.
- **Geopolitical (#4):** If U.S.–China tension escalates, it could lead to new export bans or even disrupt TSMC in Taiwan, stopping NVIDIA's supply chain.

3.3 Heat Maps

Earlier, I created a heat map showed that Risks #1 and #4 clusters in the top-right zone (high chance, high impact). That confirmed we should do more analysis for those two risks.

Figure 1. NVIDIA Risks Heat Map (1–9 Scale)

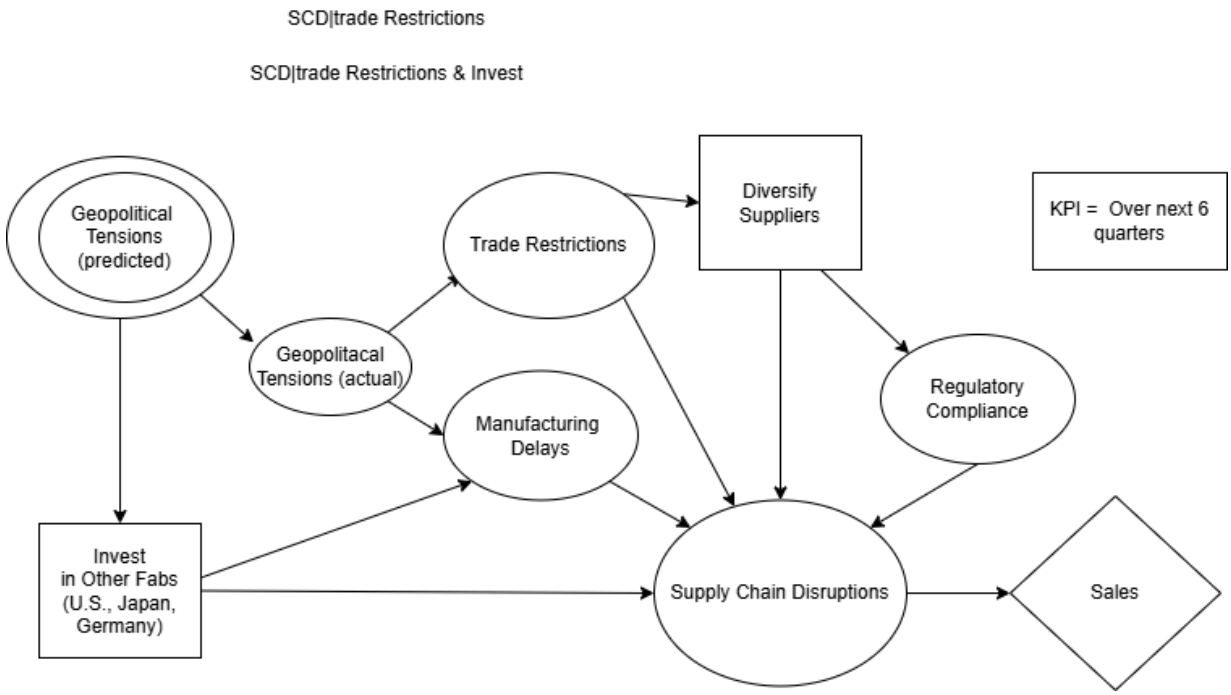


4. Risk Scenario Modeling & Treatment Plans

Now that we know which risks are most serious, we can use influence diagrams, scenario trees, and specific treatments to tackle them.

4.1 Influence Diagram (Focus on Geopolitical Tensions)

Figure 2. NVIDIA Influence Diagram



- **KRI:** Geopolitical instability affecting semiconductor supply.
- **Response Trigger:** Trade restrictions, geopolitical escalation..etc.
- **Mitigation Strategy:** Invest in other fabs and diversified suppliers.
- **KPI Impact:** Protect long-term sales and supply stability.

4.2 Scenario Tree

In this analysis, geopolitical tensions act as the Key Risk Indicator, influencing NVIDIA's strategic decisions to mitigate risk. Based on the influence diagram structure, two major decisions are:

1. **Invest in manufacturing fabs in other regions (e.g., U.S., Japan, Germany)**
2. **Increase supplier diversification**

Supplier Diversification Path

If NVIDIA chooses to **diversify suppliers**, its business performance may be influenced by Regulatory Compliance. There are three possible outcomes:

- **Low Impact:** No significant regulatory changes, allowing NVIDIA to maintain **higher quarterly sales ($16.0B, +5%)**
- **Medium Impact:** Some regulatory restrictions are imposed, keeping **sales at the baseline ($15.2B, 0%)**
- **High Impact:** Strict new regulations delay shipments, causing **a sales decline ($14.5B, -5%)**

These sales numbers are aligned with NVIDIA's 2024 10-K report, which can make the projected values close to realistic.

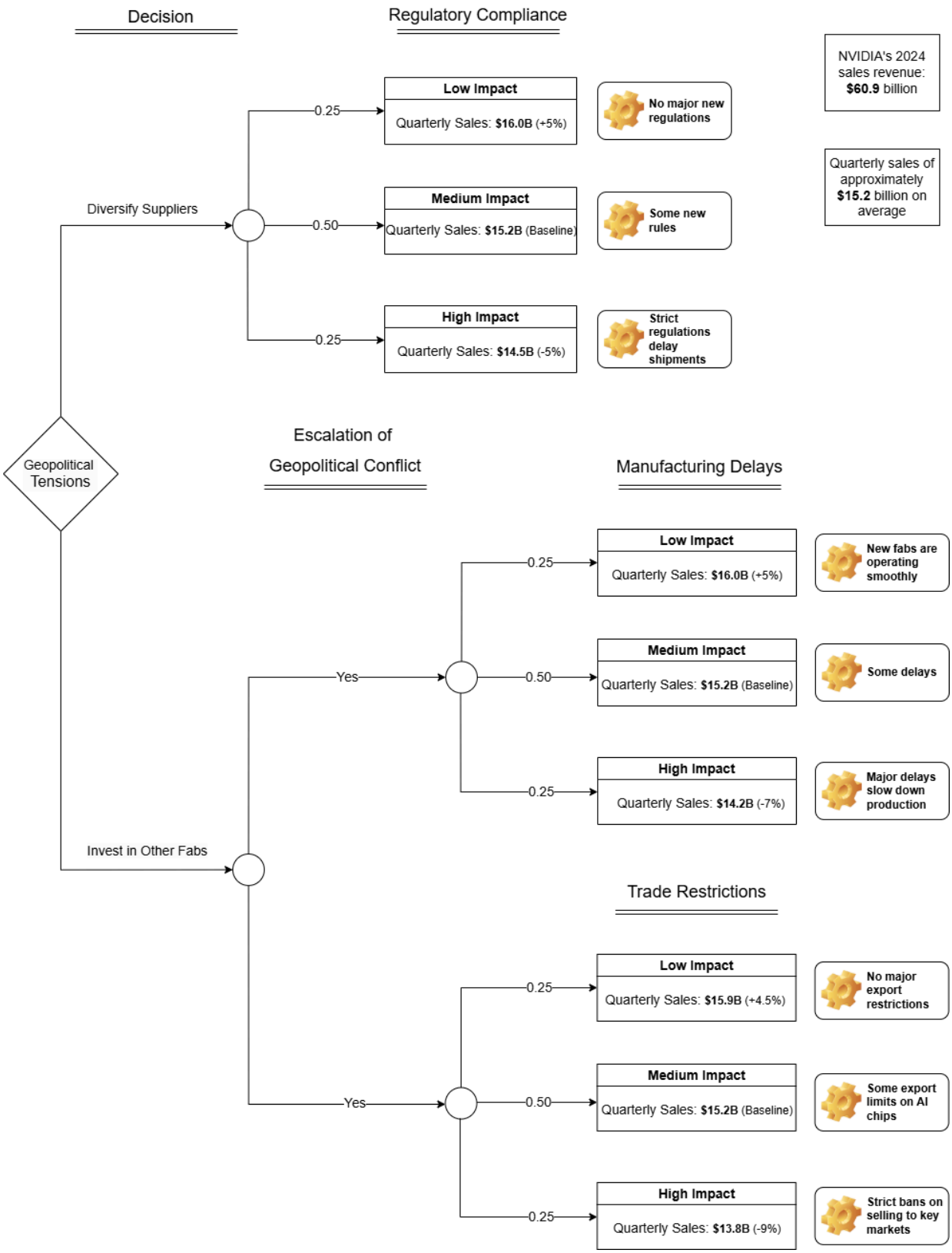
Investment in Other Fabs Path

If NVIDIA chooses to **invest in fabs outside Taiwan**, there are three possible outcomes:

- **Manufacturing Delays:** The speed of fab construction and production readiness can vary:
    - **Low Impact:** Smooth operations, **quarterly sales rise to $16.0B (+5%)**
    - **Medium Impact:** Minor production delays, **sales remain at $15.2B (0%)**
    - **High Impact:** Major delays slow production, **sales drop to $14.2B (-7%)**
- **Trade Restrictions:** Export policies and geopolitical shifts could restrict NVIDIA's ability to sell AI chips in key markets:
    - **Low Impact:** No major export restrictions, **sales rise to $15.9B (+4.5%)**

- **Medium Impact:** Some restrictions on AI chip exports, **sales remain at $15.2B (0%)**
- **High Impact:** Strict bans on selling to key markets, **sales drop to $13.8B (-9%)**

Figure 3. NVIDIA Scenario Tree Diagram



4.3 Detailed Treatment Plans

(A) Supply Chain Disruptions (#1, Score 56, High)

- **Mitigation**:
  - Sign deals with other foundries (Samsung/Intel).
  - Expand chip manufacturing in Phoenix, Japan, Germany.
  - Keep extra inventory of essential chips.
- **Contingency**:
  - If TSMC backlog hits over 4 weeks, shift partial orders.
  - If shipping times keep growing, top supply chain managers intervene.

(B) Geopolitical Tensions (#4, Score 63, High)

- **Mitigation**:
  - Track political events weekly, set up alternate shipping routes, invest in non-Taiwan fabs.
- **Contingency**:
  - If new export bans appear, quickly move production to safer regions.
  - Work with the legal team if there's any sudden policy shift.

(C) Medium Risks

- **Export Restrictions (#2)**: Partnerships in safe markets, local product versions.
- **Strong Competition (#3)**: More frequent GPU releases, deeper R&D.
- **Low-Cost AI (#5)**: Offer budget-friendly AI solutions as well as premium lines.
- **Market Volatility (#6)**: Transparent investor relations, strong financial planning.

5. Quantitative Risk Analysis

5.1 Data Sources

- **NVIDIA 2024 10-K**: Baseline revenue ~$15.2B/quarter.
- **News Articles**: For TSMC capacity usage or possible new export laws.
- **Assumptions**: I used approximate numbers for scenario building.

5.2 Monte Carlo Simulation (Summary)

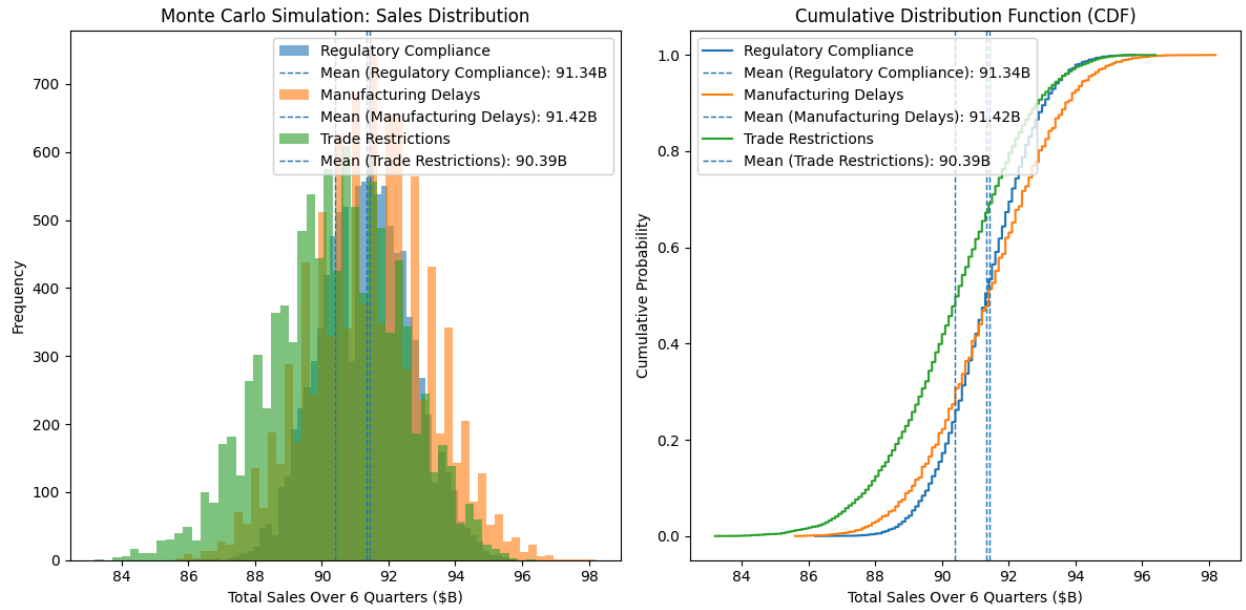
I used a Monte Carlo technique to test how changes in supply or new political rules might affect total revenue over six quarters:

- **Regulatory Compliance**: Highest average sales (~$91.34B).
- **Manufacturing Delays**: More uncertainty (~$91.42B average).
- **Trade Restrictions**: Lower average (~$90.39B) but bigger risk of a worst-case drop.

5.3 Cumulative Distribution (CDF)

We made a CDF for trade restrictions, showing a big tail on the downside, meaning a higher chance of worst-case dips in revenue. If everything is calm (Regulatory Compliance), the range is narrower and safer.

Figure 4. NVIDIA Cumulative Distribution Function (CDF)



6. Risk Management Decision Making & Key Risk Indicators

6.1 Decision Making

We used risk scoring, heat maps, scenario trees, and partial simulations. This lets management decide how to handle crucial issues, like if building new fabs is worth the cost or if we should keep bigger orders at TSMC.

6.2 Key Risk Indicators (KRIs)

- **Supply Chain (#1):**
  - TSMC capacity usage (if >85%, shift some orders).
  - Lead times over 4 weeks = immediate escalation.
- **Geopolitical (#4):**
  - Export ban rumors or official announcements = fast expansion in the U.S.

- Tensions near Taiwan (military or legal news) = accelerate local production builds.

If any of these KRI thresholds are hit, the relevant risk owner must act quickly.

7. Continuous Risk Management & Communication

7.1 Ongoing Updates

This plan is not static. We need ongoing:

- **Weekly/Monthly Reports**: Summaries of TSMC backlog, competitor moves, or any sign of new chip bans.
- **Real-Time Alerts**: If a KRI hits the danger threshold, jump to a contingency.

7.2 Organizational Communication

- **Frontline Staff**: They see shipping or production issues first, so they must speak up.
- **Finance/Investor Relations**: Must understand if risk changes will affect quarterly earnings or stock price.
- **Executives**: Approve or deny major investments (like co-developing more fabs with other foundries).

7.3 Management's Role

- Ensure budgets are available for new capacity or partnerships.
- Host risk review meetings so each risk owner can give updates.

- Keep a good **no-blame** culture so employees feel safe reporting concerns early, don't make Boeing's mistake.

8. Conclusion

NVIDIA is a leader in GPUs and AI, so it has a ton of upside but also faces real threats. Two top ones are:

- **Supply Chain Disruptions (#1)**: If TSMC can't deliver, NVIDIA might miss big sales targets.
- **Geopolitical Tensions (#4)**: If a U.S.-China conflict escalates, it could block exports or disrupt TSMC in Taiwan.

Finally, by **diversifying suppliers**, **investing in stable locations**, and **monitoring government moves**, Nvidia can reduce these risks. The other medium-priority risks should still be watched in case they become bigger.

9. Reference

1. **NVIDIA. (2024) Annual Report (Form 10-K)** *(For Nvidia's Information Research)***:** https://s201.q4cdn.com/141608511/files/doc_financials/2024/ar/NVIDIA-2024-Annual-Report.pdf
2. **National Aeronautics and Space Administration. (2011) NASA Risk Management Handbook** *(For General Concepts Research)***:** (NASA) NASA Risk Management Handbook.pdf

3. **Business Model Analyst. (2024) NVIDIA SWOT analysis** *(For Risk Register Research)***:** https://businessmodelanalyst.com/nvidia-swot-analysis/?srsltid=AfmBOooHFwwb0zyldwopoXCJY-rsTELb55X-HKFHaZndwraQMRVlHlbI#Dominance_in_GPU_Technology
4. **Peter McNamee & John Celona (2008) Decision Analysis for the Professional** *(For Influence and Scenario Tree Diagram Research)***:** https://smartorg.com/decision-analysis-for-the-professional/
5. **CNBC. (2024). TSMC says the first advanced U.S. chip plant 'dang near back' is on schedule. Here's an inside look at the Arizona fab** *(For Risk Treatment and Response Plan Research)***:** https://www.cnbc.com/2024/12/13/inside-tsmcs-new-chip-fab-where-apple-will-make-chips-in-the-us-.html
6. **Wccftech. (2024). NVIDIA Once Again Teases Potential Deal With Intel: Says Would Love To Have A Third Foundry Partner Besides TSMC & Samsung** *(For Risk Treatment and Response Plan Research)***:** https://wccftech.com/nvidia-teases-potential-intel-deal-says-would-love-third-foundry-partner-besides-tsmc-samsung/
7. **Hinrich Foundation. (2024). TSMC in Japan: Geopolitics, inward investment, and semiconductors** *(For Risk Treatment and Response Plan Research)***:** https://www.hinrichfoundation.com/research/wp/tech/tsmc-in-japan/
8. **Nasdaq. (2024). Escalating Geopolitical Tensions May Impact Nvidia and Tech Stocks** *(For Risk Treatment and Response Plan Research)***:**

https://www.nasdaq.com/articles/escalating-geopolitical-tensions-may-impact-nvidia-and-tech-stocks

9. **Reuters. (2024). Exclusive: TSMC in talks with Nvidia for AI chip production in Arizona, sources say** *(For Risk Treatment and Response Plan Research)***:** https://www.reuters.com/technology/tsmc-talks-with-nvidia-ai-chip-production-arizona-sources-say-2024-12-05/
10. **Restack. (2025) NVIDIA AI supply chain Challenges** *(For Risk Identification and Analysis)***:** https://www.restack.io/p/nvidia-ai-supply-chain-answer-cat-ai